

Supplementary Material for “Affordable Real Style Transfer: Training Wavelet-Decoupled Rectified Flow on an RTX 3060 in Minutes”

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A Method Details

This supplement expands the final WD-VF model, the proofs used in the main paper, and the supporting controls referenced in Section 4.

A.1 Problem Setup

Let $x_c \in \mathbb{R}^{H \times W \times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We encode x_c with a frozen SDXL VAE,

$$z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32},$$

sample a target latent $z_1 \sim p_s$, and use the rectified-flow path

$$z_t = (1 - t)z_0 + tz_1, \quad u_t = z_1 - z_0, \quad t \sim \mathcal{U}([0, 1]).$$

The learned generator integrates a style-conditioned velocity field from $t = 0$ to $t = 1$ and decodes the final latent with the frozen VAE decoder.

A.2 Haar Coordinates and Subband Separation

Lemma (Wavelet isometry). *The single-level 2D Haar transform $\mathcal{W}$ is orthogonal. For any latent z with subbands $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$,*

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2.$$

Proof. The one-dimensional Haar matrix is orthogonal. The two-dimensional transform is its separable Kronecker product, which remains orthogonal. Parseval’s identity then gives the stated energy decomposition. $\square$

Proposition (Content perturbation bound). *Let $z = z_{\text{cnt}} + z_{\text{sty}}$ and write $\mathcal{W}(z_{\text{cnt}}) = (\ell_{\text{cnt}}, h_{\text{cnt}})$. If*

$$\|h_{\text{cnt}}\|_F^2 \leq \epsilon \|z_{\text{cnt}}\|_F^2 \quad \text{for some } \epsilon \in [0, 1],$$

then any high-frequency-only map $T_\ell(\ell, h) = (\ell, \phi(h))$ satisfies

$$|\|T_\ell(z) - z\|_F^2 - \|\phi(h) - h\|_F^2| \leq \epsilon \|z_{\text{cnt}}\|_F^2.$$

Proof. By Haar orthogonality, all cross terms between ℓ and h vanish. The only content perturbation left is the high-frequency residue already present in h_{cnt} . $\square$

Definition (Wavelet coordinate split). *We use the LL subband ℓ as the low-frequency coordinate and the three high-frequency subbands (h_1, h_2, h_3) as the high-frequency coordinates. Equivalently, the projection map is $\pi(z) = \ell$.*

A.3 Implemented WD-VF Model

Let

$$\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t}), \quad \mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3).$$

The final WD-VF model predicts three velocity heads (v_ℓ, v_1, v_2) . The h_3 band remains in the state and is stylized at the endpoint, but it has no learned velocity head.

The implemented supervised objective is

$$\mathcal{L}_{\text{impl}} = \mathbb{E}_t [\lambda_\ell \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2], \quad (1)$$

with $\lambda_\ell = 0.3$. The LL term is therefore de-weighted, not removed.

High-frequency query routing is stochastic during training and deterministic at evaluation. Let

$$q_{\text{routed}} = Q[h_{1,t}; h_{2,t}; h_{3,t}], \quad q_{\text{full}} = Q[\ell_t; h_{1,t}; h_{2,t}; h_{3,t}].$$

During training,

$$q_t = \begin{cases} q_{\text{routed}}, & B = 1, \\ q_{\text{full}}, & B = 0, \end{cases} \quad B \sim \text{Bernoulli}(0.8),$$

while evaluation always uses q_{routed} .

A.4 Style-Agnostic Basin in Euclidean Coordinates

Definition (Style-independent manifold). *The style-independent manifold is*

$$\mathcal{M}_0 = \{\theta : \tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) \text{ for all } z, t, s\},$$

with $\bar{v}_\theta(z, t) = \mathbb{E}_s[\tilde{v}_\theta(z, t, s)]$.

Theorem (Content-dominant Euclidean basin). *Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions:*

$$\nabla_\perp \mathcal{L}_{\text{FM}}(\theta_0) = 0, \quad \nabla_\perp \mathcal{L}_{\text{sty}}(\theta_0) = 0.$$

Suppose the normal curvatures satisfy

$$\nabla_\perp^2 \mathcal{L}_{\text{FM}}(\theta_0) \succeq \mu I, \quad \|\nabla_\perp^2 \mathcal{L}_{\text{sty}}(\theta_0)\|_{\text{op}} \leq \nu.$$

If $w_f \mu > w_s \nu$, then θ_0 is a strict local minimum of $\mathcal{L}_{\text{euc}}$ restricted to $T_{\theta_0}^\perp \mathcal{M}_0$.

Proof. For any normal perturbation $\delta \in T_{\theta_0}^\perp \mathcal{M}_0$, the stationarity conditions remove all first-order terms. A second-order expansion gives

$$\Delta \mathcal{L}_{\text{euc}}(\delta) = \frac{1}{2} \delta^\top \left(w_f \nabla_\perp^2 \mathcal{L}_{\text{FM}}(\theta_0) + w_s \nabla_\perp^2 \mathcal{L}_{\text{sty}}(\theta_0) \right) \delta + o(\|\delta\|^2).$$

Using the curvature bounds,

$$\Delta \mathcal{L}_{\text{euc}}(\delta) \geq \frac{1}{2} (w_f \mu - w_s \nu) \|\delta\|^2 + o(\|\delta\|^2).$$

The coefficient of $\|\delta\|^2$ is positive, so sufficiently small nonzero perturbations increase the objective. $\square$

Style-amplitude diagnostic. For the Euclidean baseline we also fit the descriptive decomposition

$$\tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) + \kappa(\theta) \Delta v_\theta(z, t, s), \quad \mathbb{E}_s[\Delta v_\theta(z, t, s)] = 0,$$

and measure $\kappa \approx 0.016$. This diagnostic complements Theorem 1 by showing that the style-conditioned component is numerically small in the Euclidean collapsed solution.

A.5 Idealized High-Frequency Query Limit

Theorem (Subband-decoupled supervised objective). *In the idealized high-frequency-query limit, $\lambda_\ell = 0$, the high-frequency query is used deterministically, and the model has no HH velocity head. The supervised objective then reduces to*

$$\mathcal{L}_{\text{ideal}} = \mathbb{E}_t [\|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2].$$

Proof. Wavelet orthogonality splits the residual norm by subband. Setting $\lambda_\ell = 0$ removes the LL residual, and the absence of an HH head removes the h_3 residual. Deterministic high-frequency queries eliminate LL from the style-query path. Only the h_1 and h_2 transport terms remain. $\square$

This theorem isolates the mechanism behind WD-VF. The final model keeps the same geometry with $\lambda_\ell = 0.3$, high-frequency queries sampled with probability 0.8 during training, and LL untouched only at the endpoint stylization stage.

A.6 Endpoint Subband Alignment

Let the integrated endpoint satisfy

$$\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3),$$

and let a reference style latent satisfy

$$\mathcal{W}(z_1^*) = (\ell^*, h_1^*, h_2^*, h_3^*).$$

For $i \in \{1, 2, 3\}$ define the whitening-and-coloring map

$$T_i(a) = \Sigma_i^{*1/2} \hat{\Sigma}_i^{-1/2} (a - \hat{\mu}_i) + \mu_i^*.$$

The practical endpoint update is

$$\hat{h}_i^+ = (1 - \alpha_i) \hat{h}_i + \alpha_i T_i(\hat{h}_i), \quad (\alpha_1, \alpha_2, \alpha_3) = (0.3, 0.3, 0.5),$$

while $\hat{\ell}^+ = \hat{\ell}$.

Proposition (Fiber-preserving endpoint map). *The endpoint update leaves the base coordinate fixed:*

$$\pi(\hat{z}_1^+) = \pi(\hat{z}_1) = \hat{\ell}.$$

Proof. Each T_i acts only on a high-frequency subband, and the practical update sets $\hat{\ell}^+ = \hat{\ell}$. Therefore the projection π is unchanged. $\square$

Proposition (α -identifiability under endpoint injection). *If a blend of strength α is applied at every step for n steps, the residual content fraction is $r_n(\alpha) = (1 - \alpha)^n$. Endpoint-only injection corresponds to $n = 1$, for which $r_1(\alpha) = 1 - \alpha$ is affine and preserves blend identifiability.*

Proof. Each injection preserves a fraction $1 - \alpha$ of the current content residue, so composing n identical injections gives $r_n(\alpha) = (1 - \alpha)^n$. The case $n = 1$ is affine in α , while $n \gg 1$ compresses the same control range exponentially. $\square$

A.7 Architecture and Hyperparameters

B Additional Experiments

B.1 Component Ablation

B.2 Pixel-Space and Resolution Controls

The control matrix shows two effects. First, latent-space transport already dominates matched pixel-space flow: even the LPIPS-optimal latent256 checkpoint improves over pixel256 by +0.0121 CLIP-S and -0.2936 LPIPS. Second, the full 512×512 WD-VF model improves further over the latent256 control. The main paper therefore benefits from both ingredients: the latent semantic prior and the wavelet-decoupled transport geometry.

Table 1: Final WD-VF model hyperparameters.

Parameter	Value
Latent space	SDXL VAE, $4 \times 32 \times 32$
Wavelet basis	single-level Haar DWT
Backbone	4 residual blocks, width 64
Style memory	learnable per-domain bank, 256 tokens $\times$ 64 dims
Velocity heads	learned 3×3 heads for LL/LH/HL; no HH head
Loss weights	$(w_{LL}, w_{LH}, w_{HL}) = (0.3, 1.0, 1.0)$
Training route	high-frequency query with probability 0.8
Evaluation route	high-frequency query always on
Endpoint stylization	per-subband WCT, scales (0.0, 0.3, 0.3, 0.5) for LL/LH/HL/HH
ODE solver	Heun, 8 steps, cosine schedule
Optimizer	AdamW, learning rate 2×10^{-4} , weight decay 10^{-4}
Batch size	24
Epochs	5
Hardware	single RTX 3060 (12 GB)
Trainable parameters	903,248
Training time	3.08 min end to end
Evaluation time	97 s on 750 pairs

B.3 Parameter-Efficient Style Expansion

Optimizing only the style-conditioning parameters preserves the closed Distinct5 result and also supports an 8-style expansion without full retraining. We therefore report style expansion as a separate style-expansion protocol rather than merging it into the closed 5-style table.

C Additional Experimental Details

C.1 Dataset Details

We use Distinct5-WikiArt-512: Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e. Each domain has 3,600 training images and 150 test images. The benchmark was selected to have low IDT CLIP-S, so it emphasizes genuinely difficult style moves rather than near-identity targets.

C.2 Evaluation Protocol

The main paper reports CLIP-S and LPIPS on a unified 750-pair protocol: 5 source domains $\times$ 5 target domains $\times$ 30 source images, including the 150 identity pairs needed for IDT calibration. The identity baseline gives CLIP-S 0.6933 and LPIPS 0.0000. All baseline rows in the main paper table are recomputed under this protocol.

C.3 Baseline Training Details

All trainable baselines are trained to convergence on the same Distinct5 training set. The measured training times are 322.6 minutes for CUT, 39.5 minutes for SaMST, and 436 minutes for SaMam. WD-VF trains in 3.08 minutes on a single RTX 3060 (12 GB).

D Observed Boundary Cases

D.1 Minimalism Target

Minimalism is largely a low-frequency palette problem. Because WD-VF keeps LL lightly supervised during transport and leaves LL untouched at endpoint stylization, the model shifts palette more conservatively than it transfers texture-heavy domains. This target family therefore shows the highest LPIPS among the five domains.

Table 2: Component ablation relative to the final WD-VF model.

Variant	CLIP-S	LPIPS	Δ CLIP-S	Note
WD-VF (full)	0.7213	0.2868	—	baseline
<i>Wavelet basis</i>				
No DWT (Euclidean)	0.7117	0.2994	−0.0096	content-style conflict returns
Daubechies-2 (vs. Haar)	0.7258	0.3288	+0.0045	stronger style, weaker fidelity
<i>Base handling</i>				
Lock LL at inference	0.7174	0.3372	−0.0039	removes useful tone drift
$w_{LL} = 1.0$ at train	0.7174	0.2774	−0.0039	content term becomes too strong
Train HH head	0.7213	0.2868	± 0.0001	negligible gain
<i>Endpoint stylization</i>				
Per-step AdaIN	0.7361	0.3843	+0.0148	repeated over-stylization
AdaIN (diagonal) vs. WCT	0.7266	0.3402	+0.0053	weaker cross-channel match
Inject LL too	0.7215	0.3856	+0.0002	global color drift
<i>Routing</i>				
$p = 0.0$ (Euclidean train)	0.7061	0.2606	−0.0152	train-test mismatch
$p = 0.5$	0.7083	0.2480	−0.0130	high-frequency path under-trained
$p = 1.0$ (always DWT)	0.7226	0.3068	+0.0013	slightly higher style, worse LPIPS
<i>Solver</i>				
Euler (vs. Heun)	0.7210	0.3129	−0.0003	lower solver accuracy
RK4 (vs. Heun)	0.7265	0.3235	+0.0052	extra cost, no operating-point gain
<i>Capacity</i>				
depth = 2 (vs. 4)	0.7172	0.2705	−0.0041	lower capacity
dim = 32 (vs. 64)	0.7153	0.2705	−0.0060	narrower backbone
gate = 0 (no gate)	0.7080	0.2641	−0.0133	weaker style control

Table 3: Pixel-space, latent-space, and resolution controls.

Model	Space	Resolution	CLIP-S	LPIPS
pixel256 e3	pixel	256×256	0.6960	0.5317
latent256 e2	latent	256×256	0.7081	0.2381
latent256 e10	latent	256×256	0.7168	0.3125
WD-VF (main)	latent	512×512	0.7213	0.2868

D.2 High-Frequency Content Overwrite

Style is injected directly into LH/HL/HH. When the content image itself carries dense high-frequency structure, such as text, fine line work, or repetitive patterns, endpoint stylization can overwrite part of that structure with target-domain brushwork statistics.

D.3 Domain-Level Rather Than Exemplar-Level Style

WD-VF transfers the style distribution of a target domain through its style-memory bank. It does not copy the unique brushwork of a single reference painting. Exemplar-level transfer would require a different inference contract and a different benchmark.

Table 4: Parameter-efficient style adaptation results.

Setting	all-pairs CLIP-S	all-pairs LPIPS	transfer CLIP-S	transfer LPIPS
Frozen-backbone Distinct5 update	0.7218	0.3020	0.6919	—
8-style extension (50 images per new style)	—	—	0.7039	0.2555